

```
import pandas as pd
from statsmodels.formula.api import ols
```

```
TPDiv = pd.read_excel ('/content/TPDiv')
```

```
print(TPDiv.columns)
```

```
Index(['HEDG', 'HEDG_CVARB', 'HEDG_EVDRV', 'HEDG_EQNTR', 'HEDG_FIARB',
      'HEDG_LOSH0', 'HEDG_EMMKT', 'HEDG_GLMAC', 'HEDG_MGFUT', 'HEDG_DEDSH',
      'SnP', 'SnP_Up', 'SnPL1_Up', 'SnPL2_Up', 'SnPL3_Up', 'SnP_Dn',
      'SnPL1_Dn', 'SnPL2_Dn', 'SnPL3_Dn'],
      dtype='object')
```

```
Ex5R3 = ols('HEDG_EVDRV ~ SnP_Up + SnPL1_Up + SnPL2_Up + SnPL3_Up + SnP_Dn
```

```
TPDivfit=Ex5R3.fit(cov_type='nonrobust')
TPDivfit.summary()
```

```
OLS Regression Results
```

Dep. Variable:	HEDG_EVDRV	R-squared:	0.629
Model:	OLS	Adj. R-squared:	0.588
Method:	Least Squares	F-statistic:	15.25
Date:	Sun, 11 May 2025	Prob (F-statistic):	7.57e-13
Time:	10:34:16	Log-Likelihood:	245.46
No. Observations:	81	AIC:	-472.9
Df Residuals:	72	BIC:	-451.4
Df Model:	8		

Covariance Type: nonrobust

	coef	std err	t	P> t	[0.025	0.975]
Intercept	0.0101	0.004	2.785	0.007	0.003	0.017
SnP_Up	0.0328	0.072	0.458	0.648	-0.110	0.175
SnPL1_Up	0.0976	0.071	1.371	0.175	-0.044	0.240
SnPL2_Up	0.0931	0.069	1.342	0.184	-0.045	0.231
SnPL3_Up	0.0474	0.069	0.687	0.494	-0.090	0.185
SnP_Dn	0.6143	0.071	8.629	0.000	0.472	0.756
SnPL1_Dn	0.2363	0.074	3.214	0.002	0.090	0.383
SnPL2_Dn	0.0491	0.076	0.643	0.522	-0.103	0.201
SnPL3_Dn	0.0369	0.077	0.481	0.632	-0.116	0.190

Omnibus: 2.598 **Durbin-Watson:** 1.301

Prob(Omnibus): 0.273 **Jarque-Bera (JB):** 1.942

Skew: -0.231 **Prob(JB):** 0.379

Kurtosis: 3.601 **Cond. No.** 74.6

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

